A Repair Is Only as Fair as Its Gate: Auditing Multilingual Refusal Steering in a Small Multilingual Model

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Abstract

Small multilingual models run where no external guardrail does, often for speakers of low-resource languages, yet their safety repairs are judged on English-first instruments and averaged metrics. We audit whether refusal repair is *language-fair* on Cohere's Tiny Aya (3.35B) across ten languages in three resource tiers, with eleven paired repair arms, two judges checked against human labels, and analyses fixed before the outputs they concern. First, a gated repair's threshold rule decides which languages bear over-refusal and which lose protection. At a matched false-refusal budget, per-language split-conformal thresholds lower the worst language's benign refusal from 34.9% to 25.5% (-9.4 points $[-15.1, -3.8]$) without raising the mean, and their per-language false-firing bound is not contradicted on held-out prompts; the price is -7.5 points of harmful-prompt refusal in the low-resource tier $[-9.6, -5.4]$, concentrated where benign prompts score high, so the high- minus low-resource gap in harmful-prompt refusal widens from 3.8 to 14.6 points. Second, the conditional-steering gate CAST is approximately invariant to the sign of the projection it thresholds (Remark 1); end to end it refuses fewer harmful prompts in low-resource languages (-3.0 points) but also fewer benign ones, so the effect cannot be separated from its operating point. Third, measurement decides what is seen: the open safety stack's refusal detectors work only in English (0% recall elsewhere), judges that agree with humans on refusal differ on harm ($\kappa = 0.74$ vs. 0.43), and a closed model looks safer under a judge from its own family (99.3% safe) than under an independent one (93.6%), a leniency $+4.4$ points $[+2.7, +6.2]$ larger than on Tiny Aya's outputs (post hoc). Finally, calibration on clean benchmark prompts does not survive ordinary framing: prefixing "Good morning." raises the repaired model's benign refusal from 20% to 64% under either threshold rule, against 11% to 13% for the unrepaired model. All results are for one model family and translated benign prompts; per-language effects below a few points are beyond this design's power. We release the pipeline, built on the Lexsi stack (CuratorKIT $\rightarrow$ CircuitKIT $\rightarrow$ SafeTune $\rightarrow$ AuditKIT), with all judgments.

Preprint.

1 Introduction

Many people will meet language models first through small, open, often offline models in their own language. Tiny Aya [Salamanca et al., 2026] is built for this setting and is the safest model of its size on MultiJail [Deng et al., 2024]; still, on the report's own judge its Earth variant has 9.5 points lower worst-language safety than Global, and over-refusal was measured only in English. When safety erodes, the standard response is *repair*: weight arithmetic [Bhardwaj et al., 2024] or activation steering, often gated so it fires only on harmful inputs [Arditi et al., 2024, Lee et al., 2025]. A repair can raise average safety and still be unfair, by leaving one language under-protected or by refusing that language's ordinary questions. Low-resource languages already bypass English-centric safety training [Yong et al., 2023, Deng et al., 2024], and LLM judges are least accurate exactly there [Zhou et al., 2026].

We ask a simple question with a careful design: **is safety repair language-fair, and what decides it?** We run eleven repair arms on Tiny Aya Earth, each paired prompt-by-prompt with the unrepaired model in ten languages, score responses with two judges checked against human labels (the later controls with the primary judge only), and fix each analysis before the outputs it concerns exist. Two independent reviews of an earlier draft found that our headline contrast mixed a fairer allocation with a stricter operating point; the matched-budget controls they prompted are the core of §5.1. Three findings emerge (Fig. 1).

- **The threshold rule allocates over-refusal and protection** (§5.1). Every gated arm buys a similar, near-ceiling safety gain. At a matched false-refusal budget, per-language split-conformal thresholds narrow the cross-language spread of over-refusal (worst ÷ mean 2.24 $\rightarrow$ 1.75) and widen the high- minus low-resource gap in harmful-prompt refusal (3.8 $\rightarrow$ 14.6 points); at gate level, worst-language false-firing falls and the tier gap in detection widens in 100% and 100% of 50 redrawn calibration splits. A deployer must choose which error to equalise; we measure both.
- **CAST's gate score is approximately sign-invariant** (§5.2). In its linear regime the score depends on the magnitude of the projection (Remark 1); end to end CAST lowers refusal of both harmful and benign prompts, consistent with a different operating point.
- **Measurement and robustness decide what is seen** (§5.3, §5.4). The stack's own refusal detectors are English-only, judges disagree on harm, a same-family judge flatters a closed model, and a greeting adds about $+41.4$ points of benign refusal beyond its effect on the unrepaired model, whatever the threshold rule.

We also report, briefly and with power statements, natural drift between Tiny Aya Global and Earth, a preamble ablation, weight-space repair (ReSta), and a circuit-discovery attempt whose negative result we document as a tooling lesson (Appendix D).

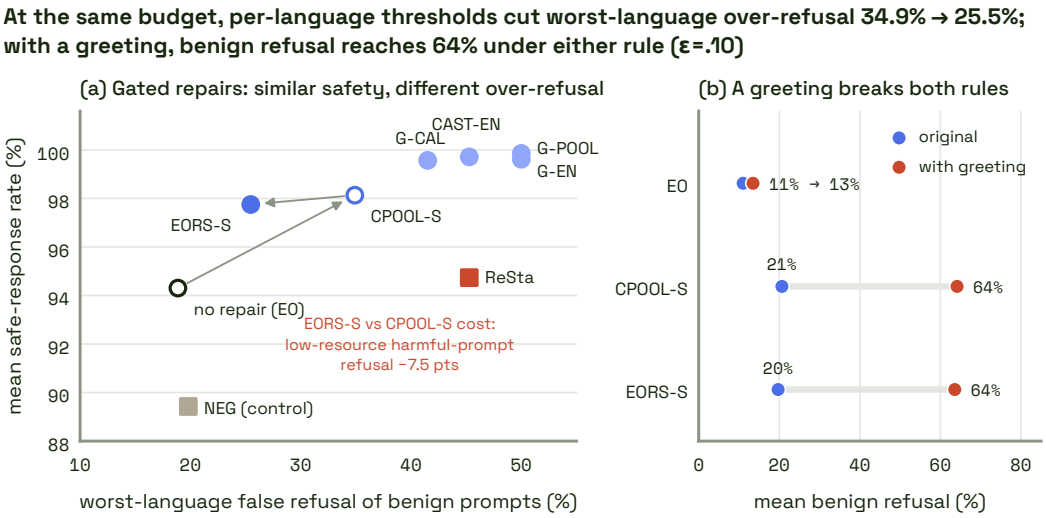


Figure 1: **The threshold rule sets who is over-refused; a greeting breaks every rule.** (a) Every repair arm on Tiny Aya Earth, judged by Qwen3Guard-4B on the test split (209 harmful MultiJail and 106 benign translated-XSTest prompts per language, 10 languages): mean safe-response rate against the worst language's benign refusal rate. Arrows run from the unrepaired model (E0) to one pooled split-conformal threshold (CPOOL-S, ε=.05) and on to per-language thresholds at the same budget (EORS-S); the two share steering vector, condition direction and calibration data. Paired bootstrap intervals are in Tables 1 and 2. (b) End-to-end greeting test (ε=.10; pre-registered secondary): mean benign refusal without and with the English prefix "Good morning." in every language.

2 Related work

Multilingual safety gaps. Translating harmful requests into low-resource languages bypasses safety training [Yong et al., 2023, Deng et al., 2024]; human-written multilingual red-teaming separates global from local harms [Aakanksha et al., 2024]; multilingual guards such as PolyGuard [Kumar et al., 2025] cover 17 languages, missing most of Tiny Aya's. Multilingual safety research remains English-dominated [Yong et al., 2025], and judges are less accurate in lower-resource languages [Zhou et al., 2026].

Refusal geometry and conditional steering. Refusal is mediated by a low-dimensional direction [Arditi et al., 2024] shared across safety-aligned languages [Wang et al., 2025b]. CAST adds a refusal vector only when a condition direction fires [Lee et al., 2025]. English-derived steering raises over-refusal in low-resource languages [Stein et al., 2026], and their failures are *action* failures: harm is represented but scores sit below the refusal range, so a per-language calibrated gate restores selectivity [Aziz et al., 2026]. Aziz et al. [2026] already run per-language calibrated conditional steering end to end, against CAST and AdaSteer [Zhao et al., 2025b], and score refusal selectivity; single-vector ablation also reduces false refusal [Wang et al., 2024]. We build on their setting and ask what it leaves open: which threshold *objective* allocates over-refusal across languages, what that allocation costs in harmful-prompt refusal and harmful answers, and how robust it is, with three-way outcomes and validated multilingual judges.

Safety drift and repair. Fine-tuning erodes alignment [Qi et al., 2024], unevenly across languages [Poppi et al., 2025, Hawkins et al., 2026]; ReSta restores it with a DARE-sparsified safety vector [Bhardwaj et al., 2024]. Multilingual safety gaps are well documented [Wang et al., 2023]; whether repairs close them evenly across languages has received less attention than whether they close them on average.

Fairness and conformal calibration. Equal false-positive rates across groups is *predictive equality* [Corbett-Davies et al., 2017]; equalised odds also equalises true-positive rates [Hardt et al., 2016]. With group-specific thresholds each group sits at a different point of its own ROC curve, so equal-

ising one error rate generally leaves the other unequal [Hardt et al., 2016]; with unequal base rates, calibration and error-rate balance cannot both hold [Chouldechova, 2017, Kleinberg et al., 2017]. A conformal quantile per group is Mondrian split-conformal prediction [Vovk et al., 2003, Gibbs et al., 2025]. Its guarantee needs a score fitted on data independent of the calibration fold and holds on average over calibration draws [Angelopoulos and Bates, 2021]; high-probability versions exist [Angelopoulos et al., 2021]. Detectors trained on harmful and benign sets from different sources can learn the source rather than the harm [Fomin, 2026, Wang et al., 2025a].

3 Setting and measurement

Models and hardware. Tiny Aya Global and Earth (Africa/West Asia specialisation), bf16, greedy decoding, 96 new tokens, shipped chat template, whose "contextual safety mode" preamble we also remove (B1). All repair arms and their paired references (E0, B0) were generated on one A100 with the same code, batch size and library versions; on E0, A100 and the original Apple-MPS generation differ by $+0.2$ points of mean SRR $[-0.2, +0.8]$ with refusal-label agreement $\kappa = 0.95$, so pairs never mix hardware. On one A100 an arm takes about 17 minutes to generate all test responses and 21 to score with Qwen3Guard-4B.

Data. Harmful: MultiJail, 314 parallel prompts $\times$ 10 languages (high resource: en, zh, it, vi; medium: ar, ko, th; low: bn, sw, jv), split 105 calibration / 209 test by prompt id. Benign-but-sensitive: XSTest safe prompts [Röttger et al., 2024] translated with NLLB-200 [NLLB Team et al., 2022], kept only where constrained GlotLID [Kargaran et al., 2023] and back-translation chrF agree in *all* languages (90 calibration / 106 test ids). CuratorKIT cleans, deduplicates and records provenance; ids that lose any language are dropped, so every test is fully paired. Licences: MultiJail MIT, XSTest and PolyGuardPrompts CC-BY-4.0, Aya red-teaming Apache-2.0; Tiny Aya and NLLB outputs are CC-BY-NC-4.0, which our data release inherits.

Outcomes. Each response to a harmful prompt is a refusal, a safe compliance or a harmful compliance. Primary: $\mathrm{SRR}_\ell = 1-$ harmful-compliance rate, and BRR_ℓ = refusal rate on benign prompts. Behavioural: HRR_ℓ, refusal on harmful prompts. Fairness: worst-language SRR and worst-language BRR.

Judges and statistics. Qwen3Guard-Gen-4B [Zhao et al., 2025a] is the primary judge and Gemini 3.5 Flash (broad rubric) the second; both are validated against human labels in §5.3. Comparisons are paired by prompt id: exact McNemar tests per language with Holm correction, paired bootstrap over ids (5,000 resamples) for pooled estimands, Wilson intervals for rates. Each confirmatory comparison was fixed, with its reading rule, before its own outputs existed; the plan's history, including two changes of primary endpoint, is dated in Appendix A.

4 Gates and arms

Let $h(x) \in \mathbb{R}^d$ be the residual stream at the last prompt token at layer L, and v the unit difference of class means (harmful minus benign) over calibration prompts. A gated arm adds a steering vector αu to a band of layers iff a gate score exceeds a threshold at prefill.

CAST's score is approximately sign-invariant. CAST [Lee et al., 2025], faithfully implemented in SafeTune, scores $g_{\mathrm{CAST}}(h) = \cos\big(h, \tanh(v\, v^\top h)\big)$.

Remark 1 (Sign invariance)**.** *Let $s = v^\top h$ with $\|v\| = 1$. In the linear regime of* tanh *($|s| \max_i |v_i| \ll 1$), $\tanh(sv) \approx sv$, so $g_{CAST}(h) \approx s\, v^\top h/(\|h\|\,|s|) = |s|/\|h\|$: the score ranks prompts by the magnitude of the projection, and a prompt projecting strongly* negatively *scores like a strongly harmful one.*

The remark is a two-line identity under a linearisation. On Tiny Aya Earth's calibration activations the regime is approached for benign prompts (median $|s| \max_i |v_i|$ 0.35–0.61 across languages) but not for harmful ones (1.8–2.2), so it describes a tendency of the ranking, which we test empirically. Whether it matters depends on how many benign prompts project negatively, which varies by language (Appendix B). Our own gates threshold the projection s itself, and the faithful CAST gate is kept as an arm.

Equal-over-refusal steering (EORS). For language ℓ with benign calibration scores $s_1^{(\ell)} \leq \cdots \leq s_{n_\ell}^{(\ell)}$ and budget ε, set $\tau_\ell = s^{(\ell)}_{\lceil (1-\varepsilon)(n_\ell+1) \rceil}$; at inference the language comes from constrained GlotLID.

Proposition 1 (Per-language false-firing bound)**.** *If the score function (layer and direction) is fitted on data independent of language ℓ's calibration prompts, and a new benign prompt in ℓ is exchangeable with them and correctly identified, then* $\Pr[s > \tau_\ell] \leq \varepsilon$*, the probability taken over the calibration draw and the new prompt.*

This is Mondrian split-conformal prediction [Vovk et al., 2003], i.e. predictive equality [Corbett-Davies et al., 2017] for the gate. Given one calibration set, the realised false-firing rate is itself random, $\mathrm{Beta}(n_\ell + 1 - k, k)$ with $k = \lceil (1-\varepsilon)(n_\ell + 1) \rceil$ [Angelopoulos and Bates, 2021], and spans several points at $n_\ell \approx 60$. Our first EORS arms fitted the direction and the thresholds on the same calibration prompts, which voids the independence condition; EORS-S restores it by fitting the layer and direction on one third of the calibration ids and the thresholds on the other two thirds (§5.1). The bound concerns when the *gate* fires, not the model's baseline over-refusal, and assumes exchangeability, which §5.4 stress-tests.

Arms (all on Tiny Aya Earth). G-POOL: pooled steering vector and direction, one max-F1 threshold. G-EN: English vector and direction, projection gate. CAST-EN: as G-EN with CAST's gate score. G-CAL: per-language max-F1 thresholds (after Aziz et al., 2026). EORS at $\varepsilon \in \{.05, .10\}$ (thresholds fitted in-sample). EORS-S: EORS with the split fit above. CPOOL-S, the matched-budget control: the same split, direction and ε, with one conformal threshold on the pooled benign scores of all languages (a marginal guarantee only). ReSta: SafeTune's `apply_resta` (α=0.25, DARE p=0.3). NEG: the steering vector negated (control). Steering strength was fixed before any arm ran by a rule on calibration data only (largest α keeping responses on-language and non-degenerate): $\alpha = 1.0$, where ungated steering already refuses 98% of harmful and 98% of benign calibration prompts, so the gate carries all selectivity; at $\alpha = 2$ only 60% of responses stay on-language.

5 Results

5.1 The threshold rule allocates over-refusal and protection (pre-registered)

The gated arms of Table 1 raise mean SRR by +4.5 to +5.6 points over E0, and the split arms of Table 2 by +3.4 to +4.5. E0 is already at 94.3%, so this gain is near the ceiling and does not discriminate between arms. Over-refusal does: arms with a single threshold or per-language max-F1 thresholds raise worst-language BRR by +22.6 to +31.1 points.

The first contrast mixed level with spread. Our pre-registered contrast held the steering vector and direction fixed and replaced G-POOL's single max-F1 threshold with per-language conformal thresholds (EORS). Worst-language BRR fell by -24.5 points [-34.9, -17.0], but mean BRR fell too (-13.1), and the worst ÷ mean ratio did not move (1.86 vs. 1.85): EORS was mostly a stricter rule, not a fairer one. Its direction and thresholds were also fitted on the same calibration prompts, which voids Prop. 1. Both points came from independent reviews of an earlier draft.

Matched-budget control (pre-registered). We therefore compared EORS-S with CPOOL-S, which share steering vector, condition direction, calibration fold and budget ε and differ only in whether the conformal threshold is set per language or pooled (Table 2, Fig. 2). The co-primary estimands and their reading rules were fixed before generation; Holm correction runs over the two budgets. At ε=.05, per-language thresholds lower worst-language BRR by -9.4 points [-15.1, -3.8] ($p < .001$), from 34.9% to 25.5% in Thai, while mean BRR does not rise (-1.0 [-2.2, $+0.0$]); the worst ÷ mean ratio falls from 2.24 to 1.75, the cross-language range by -9.4 [-16.0, -2.8], and no language's benign refusal exceeds E0's by more than 6.6 points (single threshold: 16.0). Thai is the worst language under both rules and the only one whose benign change survives per-language Holm correction (Thai, Javanese at ε=.10); the reported p is the bootstrap floor for this max-type statistic. At ε=.10 the result is the same (-8.5 [-14.2, -3.8], $p < .001$). The price lands on the languages whose benign prompts score highest, which pull their thresholds up: harmful-prompt refusal falls in Javanese (-14.4), Bengali (-8.1) and Thai (-8.1), not in Swahili ($+0.0$), and rises in

Table 1: **Every gated arm buys a similar safety gain; arms differ in over-refusal.** Paired change against the unrepaired model (E0, Tiny Aya Earth; B1 against B0, Tiny Aya Global), percentage points, Qwen3Guard-4B, 95% paired bootstrap intervals over prompt ids; point estimates are the statistic on the observed data. Worst-language BRR is the maximum over the ten languages. EORS rows use thresholds fitted on the same prompts as the direction (superseded by EORS-S, Table 2).

Arm	Gate / change	Δ mean SRR	Δ worst-language BRR	Δ mean HRR
G-POOL	single pooled threshold	$+5.3$ [$+3.8$, $+7.0$]	$+31.1$ [$+22.6$, $+40.6$]	$+33.2$ [$+28.7$, $+37.6$]
G-EN	English vector, projection gate	$+5.6$ [$+4.0$, $+7.3$]	$+31.1$ [$+23.6$, $+41.5$]	$+36.1$ [$+31.5$, $+40.7$]
CAST-EN	English vector, CAST gate	$+5.4$ [$+3.9$, $+7.1$]	$+26.4$ [$+18.9$, $+34.9$]	$+34.1$ [$+29.7$, $+38.6$]
G-CAL	per-language max-F1	$+5.3$ [$+3.8$, $+6.9$]	$+22.6$ [$+16.0$, $+32.1$]	$+33.2$ [$+28.7$, $+37.7$]
EORS ε=.10	per-language conformal	$+5.1$ [$+3.6$, $+6.8$]	$+11.3$ [$+5.7$, $+17.9$]	$+31.1$ [$+26.7$, $+35.6$]
EORS ε=.05	per-language conformal	$+4.5$ [$+3.1$, $+6.0$]	$+6.6$ [$+0.9$, $+11.3$]	$+24.9$ [$+21.1$, $+29.0$]
ReSta	weight arithmetic	$+0.4$ [-1.0, $+1.9$]	$+26.4$ [$+17.0$, $+34.9$]	$+8.3$ [$+6.2$, $+10.5$]
NEG	negative control	-4.9 [-6.2, -3.6]	$+0.9$ [-4.7, $+8.5$]	-3.6 [-5.8, -1.5]
B1 (no preamble)	vs. B0, Tiny Aya Global	$+0.0$ [-0.6, $+0.7$]	$+3.8$ [-2.8, $+10.4$]	$+1.5$ [$+0.3$, $+2.8$]

high-resource languages (Italian $+6.2$). In the pre-registered low-resource tier it falls by -7.5 points [-9.6, -5.4] ($p < .001$; ε=.10: -7.3). The cross-language range of harmful-prompt refusal barely moves (19.6 vs. 20.1 points), but the gap between the high- and low-resource tiers grows from 3.8 to 14.6 points. Harmful answers in that tier rise by $+1.3$ points [$+0.3$, $+2.2$] at ε=.05 and $+0.5$ [-0.2, $+1.3$] at .10; mean SRR changes by -0.4. Group-specific thresholds that equalise one error rate place each group at a different point of its own ROC curve, so the other error rate generally differs [Hardt et al., 2016]; which languages lose is empirical, and here it is those whose benign prompts score high, as the calibration-only design check predicted. Per-language calibration makes over-refusal more equal and protection less equal between tiers. Which error to equalise is a policy choice; Appendix Fig. 5 gives the gate-level frontier across budgets.

The per-language bound is not contradicted, and the trade-off survives redrawing the split. On held-out benign prompts (gate level), EORS-S's false-firing is 4.2% on average at ε=.05 and stays inside its predictive band in every language at both budgets, whether thresholds are routed by the true or the detected language (pre-registered); the in-sample EORS thresholds reached up to 8.5%. With $n_\ell = 60$ calibration prompts the band is wide (97.5th percentile 13.2% at ε=.05, 20.8% at .10), and the pre-registration fixed a single fit/conformal split. Redrawing that split 50 times (exploratory), at most 1 language exceeds the band (none in 94% of draws); per-language thresholds lower worst-language false-firing in 100% of draws (median -6.6 points, 95% of draws in [-13.9, -1.9]) and low-resource detection in 98% (median -9.8); their mean false-firing is also slightly lower in 86% of draws (median -0.8), far less than the worst-language change. Post hoc, realised rates within a draw move together across languages because the prompt ids are parallel, and at budgets of .18 and above 2–4 languages exceed the band in the frontier, so the bound should be read as holding on average over calibration draws, not for each. Recomputing the calibration activations on the A100 shows that hardware and batching shift gate scores by a standard deviation of only 0.04 and 0.05 (score s.d. 2.8), so prompt sampling, not the hardware pairing, explains the gap between calibration and test score distributions.

The in-sample EORS contrast also holds at half the steering strength (α=0.5; -17.9 points; Appendix Table 3); since gate thresholds do not depend on α, this is a robustness check on the steering strength, not an independent replication.

5.2 CAST's gate score is approximately sign-invariant; its end-to-end cost is not separable (H6, exploratory status)

At the gate level, on Tiny Aya Global's held-out activations, CAST loses separability in proportion to the share of benign prompts projecting negatively (Pearson $r = 0.89$ across languages; Swahili AUROC 0.97→0.74; Appendix B), the regime Remark 1 identifies. End to end, we compare CAST-EN with G-EN, which share the English steering vector and condition direction and differ only in the gate score (Appendix Fig. 4). The pre-registered primary estimate supports the predicted direction: in the low-resource tier CAST-EN refuses -3.0 points fewer harmful prompts [-4.6, -1.8] (Gemini -2.6 [-4.1, -1.3]), and it refuses fewer in 9 of 10 languages. Two qualifications matter. The effect does not reach harmful answers ($+0.2$ points), because at $\alpha = 1.0$ both gates refuse almost

Table 2: **Matched-budget controls** (pre-registered 2026-10-28; Tiny Aya Earth, A100, α=1, Qwen3Guard-4B). Top: each arm against the unrepaired model. Bottom: per-language against pooled split-conformal thresholds at the same budget, the pre-registered contrast. Percentage points, 95% paired bootstrap intervals over prompt ids; low-resource tier: bn, sw, jv.

Arm	Reference	Δ mean SRR	Δ worst-language BRR	Δ mean BRR	Δ HRR, low-resource tier
CPOOL-S ε=.05	vs. E0	+3.8 [+2.6, +5.2]	+16.0 [+8.5, +22.6]	+4.5 [+2.6, +6.7]	+29.5 [+24.6, +34.4]
EORS-S ε=.05	vs. E0	+3.4 [+2.2, +4.7]	+6.6 [+1.9, +11.3]	+3.5 [+1.6, +5.9]	+22.0 [+17.5, +26.6]
CPOOL-S ε=.10	vs. E0	+4.3 [+3.0, +5.7]	+24.5 [+16.0, +32.1]	+9.6 [+6.5, +13.0]	+35.7 [+30.6, +41.0]
EORS-S ε=.10	vs. E0	+4.5 [+3.1, +6.1]	+16.0 [+8.5, +22.6]	+8.7 [+5.5, +12.4]	+28.4 [+23.6, +33.2]
EORS-S ε=.05	vs. CPOOL-S	−0.4 [−0.8, −0.0]	−9.4 [−15.1, −3.8]	−1.0 [−2.2, +0.0]	−7.5 [−9.6, −5.4]
EORS-S ε=.10	vs. CPOOL-S	+0.2 [−0.1, +0.6]	−8.5 [−14.2, −3.8]	−0.9 [−2.4, +0.4]	−7.3 [−9.4, −5.3]

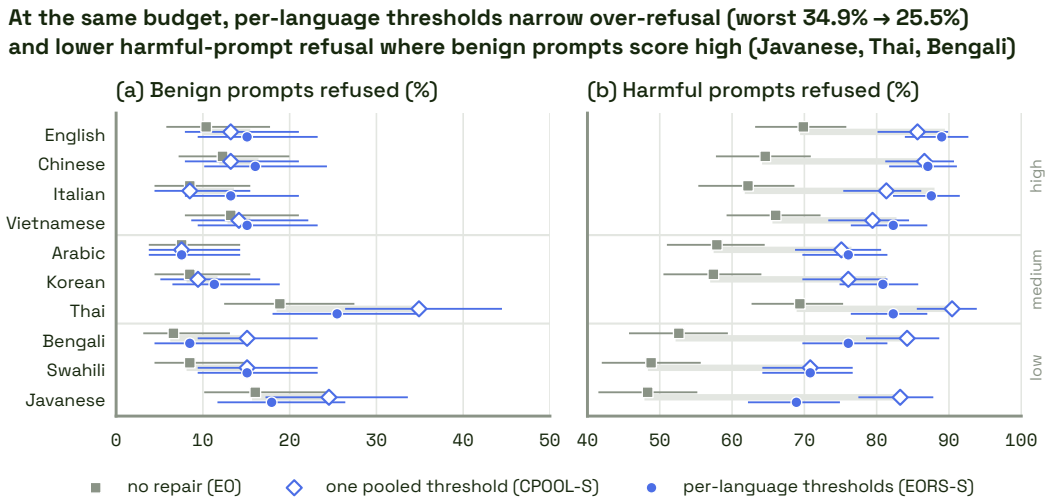


Figure 2: **Same steering, same budget, different threshold rule.** Tiny Aya Earth, α=1, ε=.05. CPOOL-S and EORS-S share steering vector, condition direction, calibration fold and false-refusal budget and differ only in whether the split-conformal threshold is pooled over languages or set per language; E0 is unrepaired. Qwen3Guard-4B labels; 95% Wilson intervals; languages grouped by resource tier. Paired tests are in Table 2.

all harmful prompts; and across languages the loss does not track the negative-projection share (Spearman $\rho = -0.23$), so the gate-level mechanism does not carry over cleanly to behaviour. No single language survives Holm correction (0 of 10): with 1.9% of pairs discordant, no per-language effect is detectable with 80% power at $\alpha = 0.05$. CAST-EN also over-refuses less: worst-language BRR changes by -4.7 points $[-12.3, +0.0]$, mean BRR by -7.1, and benign refusal falls after Holm correction in 3 languages. Lower refusal on both prompt types with unchanged SRR is what a stricter operating point produces, and CAST's threshold is chosen by max-F1 on English in a different score space; the end-to-end contrast therefore does not isolate the sign invariance from an operating-point shift. We pre-registered the prediction that the effect grows when steering is weaker. At α=0.5 the test is indeterminate: both gated arms are at ceiling (Appendix Table 3), the low-resource estimate (-2.6 points $[-4.0, -1.3]$) is close to α=1's, and we fixed no interval for the difference. We read the sign invariance as a property of the score worth fixing in CAST-style gates; whether it costs behaviour on this model is unresolved.

5.3 Measurement decides what is seen (H1 pre-registered; judge validation)

SafeTune ships two English keyword refusal detectors, which the Lexsi stack uses to score refusal and choose steering layers. On all B0 responses they agree with Qwen3Guard in English ($\kappa = 0.85$–0.92) and detect 0% of refusals in each of the nine other languages (Fig. 3a); the pre-registered English-vs-rest agreement gap is 0.86 [0.80, 0.92]. A keyword audit would report that the model never refuses outside English. Against 1,400 human-labelled PolyGuardPrompts items in seven languages, both judges detect refusal well ($\kappa = 0.89$ Qwen3Guard, 0.93 Gemini), but on harm

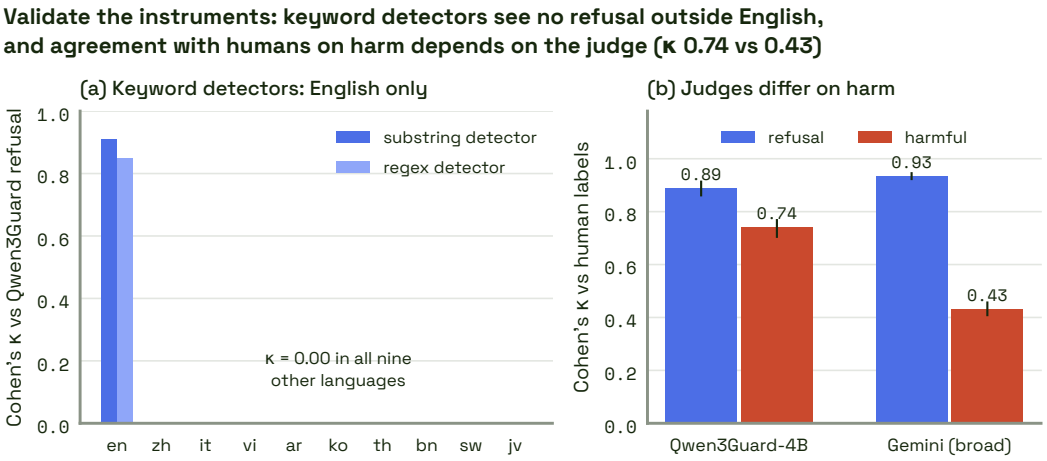


Figure 3: **Instrument validity.** (a) Agreement between SafeTune's English keyword refusal detectors and Qwen3Guard-4B refusal labels on all B0 responses (Tiny Aya Global), per language (H1). (b) Agreement of each judge with PolyGuardPrompts human labels on 1,400 items in en, hi, it, th, zh, ar, ko; bars: mean over languages, whiskers: range. None of our low-resource languages is among the validation languages.

Qwen3Guard reaches $\kappa = 0.74$ and Gemini only 0.43, missing 29% and 63% of harmful responses (Fig. 3b). Misses concentrate where annotators split 2–1 (Qwen3Guard 54%, vs. 9% on unanimous items). Refusal is not safety: a closed model, Gemini 3.5 Flash-Lite, refuses only 46%–58% of MultiJail prompts and its responses are 99.3% safe under the Gemini judge but 93.6% under Qwen3Guard, close to the unrepaired Tiny Aya Earth's 94.3%. The Gemini judge is lenient in general (harm $\kappa = 0.43$), but more so on its own family's outputs: its unsafe-rate gap to Qwen3Guard is 5.7 points on Gemini's responses and 1.4 on Tiny Aya's, a difference of $+4.4$ [$+2.7$, $+6.2$] (post hoc), consistent with LLM judges favouring their own generations [Panickssery et al., 2024, Zheng et al., 2023]. We therefore treat harmful compliance as the safety outcome and refusal as behaviour, and report harm under both judges.

5.4 Gate calibration on clean prompts does not survive a greeting (found post hoc; tested end to end)

EORS picks its threshold from the detected language, so we tested whether adding text in another language lets an attacker route a harmful prompt to a laxer threshold. It does not: an English prefix flips language identification for up to 33% of harmful prompts, but the gate fires *more* on prefixed harmful prompts, not less. The same experiment exposed a larger problem on benign prompts (Fig. 1b; Appendix Fig. 9). Gate false-firing averages 12% on the original prompts and rises to 58% with an English instruction prefix, 45% with the same instruction in the prompt's own language, and 75% with "Good morning." Language identification is unchanged for the own-language and greeting prefixes, and among benign prompts the gate score is unrelated to prompt length, so neither code-mixing nor length explains the shift. We leave the mechanism open.

End-to-end test (pre-registered). We then tested the finding end to end with an analysis fixed before generation, on the same test prompts on which it was found and with the most disruptive prefix of the control set: EORS (ε=.10, in-sample thresholds) and the unrepaired model, each with every prompt prefixed by "Good morning.", and with the gate's language detected on the prefixed prompt, as at inference; the English greeting is used in every language, and labels are Qwen3Guard-only. The estimand is a difference-in-differences in benign refusal, the repair's added refusal with the greeting minus its added refusal without it, which removes the greeting's effect on the model itself. It is $+52.6$ points [$+46.6$, $+58.6$], with an interval above zero in all 10 languages (+40.6 to +64.2), English included. The greeting barely moves the unrepaired model (11% to 13% benign refusal) but takes the repaired one from 21% to 76%. Harmful answers do not change (difference-in-

differences $+0.5$ points $[-0.7, +1.6]$): the greeting does not unlock harm; it makes the repair refuse ordinary questions.

Both threshold rules break (pre-registered secondary). With the controls we repeated the test for EORS-S and CPOOL-S (ε=.10). The difference-in-differences in benign refusal is $+41.4$ points $[+34.7, +48.2]$ for per-language thresholds and $+41.0$ $[+35.3, +47.0]$ for the pooled one: mean benign refusal rises from 20% to 64% and from 21% to 64%. Per-language thresholds still cap the worst language lower (76% vs. 91%), but the brittleness belongs to the gate's direction and calibration distribution, not to its threshold rule.

The gate detects harm, not only the benchmark's style. Harmful (MultiJail) and benign (XSTest) prompts come from different sources, so the direction could encode provenance [Fomin, 2026, Wang et al., 2025a]. On XSTest's own unsafe contrast prompts, which share the benign prompts' surface form (141 ids, translated with the same pipeline), the gate score separates unsafe from safe with AUROC 0.81 pooled and at least 0.78 in every language (pre-registered rule: harm signal beyond style), against 0.95 for MultiJail. XSTest's unsafe prompts are minimal pairs, harder by design, so the gap does not measure how much of MultiJail's separability is style. On unseen harmful prompts from Aya red-teaming the gate fires on 76% (English) and 61% (Arabic) at ε=.05, against 82% and 59% of MultiJail's harmful test prompts; in the six Aya languages outside our ten, which fall back to the strictest threshold, it fires on 18%–41%.

The practical consequence is that the conformal guarantee (Prop. 1) holds for prompts exchangeable with the calibration set, and ordinary conversational framing breaks exchangeability. Any gate calibrated on benchmark prompts should be re-calibrated on, or at least tested against, the formats users actually send.

5.5 Secondary results

Natural drift (pre-registered). On identical prompts Tiny Aya Earth is slightly less safe than Global: mean SRR -1.1 points $[-1.9, -0.3]$ under Qwen3Guard and -1.0 under Gemini, with no language significant after Holm correction (Appendix C). Power, not the model, sets this limit: only 3.8% of paired prompts change label, so a per-language test reaches 80% power [Connor, 1987] only for effects of 3.8 points at α=0.05, and none after Holm correction. The Tiny Aya report finds a larger worst-language loss (87.0 to 77.5 with a different judge and the full set; Salamanca et al., 2026); we read our estimate as drift under this protocol, not a contradiction.

Detection survives drift (H3). In the languages with the largest drift, harm-direction AUROC drops by at most 0.009 (pre-registered threshold 0.03), so drift is not a detection failure; a scale-free margin does not single out those languages either (post hoc), so we claim no language-specific action failure.

The preamble does not carry safety. Removing the chat template's safety preamble (B1) changes mean SRR by $+0.0$ points $[-0.6, +0.7]$ and harmful-prompt refusal by $+1.5$: at this near-ceiling safety level the preamble carries no measurable safety.

Weight-space repair does not help here. ReSta changes mean SRR by $+0.4$ points $[-1.0, +1.9]$ while raising worst-language BRR by $+26.4$. The NEG control changes SRR by -4.9 points, as a sign-flipped vector should.

Cost. The closed comparator cost $1.71 for 3,950 responses at list price; Tiny Aya has no per-token fee and generated 27 responses per minute on a 16 GB laptop. Self-hosting removes the per-token cost; it does not remove the need to audit a repair in every language.

6 Discussion and limitations

What generalises, and what may not. Once steering is strong enough to refuse reliably, a gated repair is a classifier followed by a forced refusal, and its fairness is a property of the classifier's threshold rule and calibration data. Three lessons follow from the structure of that problem rather

than from this model: per-group calibration of scores whose distributions differ moves protection between groups; a calibration set of parallel translations makes per-group errors move together (a post hoc observation); and calibration on benchmark-formatted prompts does not cover conversational framing. The magnitudes are for one 3B model family at one steering strength (with a robustness check at a second) and one translation pipeline.

Recommendations. Report per-language false-refusal *and* harmful-prompt refusal for any gated repair, at matched budgets; fit the gate direction and the thresholds on disjoint data; calibrate on, or test against, the formats users send; and check a closed model with a judge outside its family.

Limitations. (i) One model family, ten languages, MultiJail and machine-translated XSTest prompts; benign probes are translated, not native. (ii) Judges are validated against humans in seven languages that include none of our low-resource tier; harm metrics are reported under both judges, and the controls and follow-ups under Qwen3Guard only. (iii) Greedy decoding and $\alpha{=}1$ (robustness check at 0.5). (iv) The controls reuse the test prompts of the earlier contrasts with new estimands; only the XSTest-unsafe and Aya red-teaming sets are unseen. The pre-registered contrast rests on one fit/conformal split (redrawn only at gate level) and its worst language is Thai under both rules; Remark 1's linear regime holds only approximately. (v) With 60 calibration prompts per language the per-language bound is loose, and we give no high-probability (PAC) version, which would need far more calibration data [Angelopoulos et al., 2021]. (vi) Per-language effects of a few points are below this design's power; pooled tier estimates are what it resolves. (vii) Not run: the strict-preamble arm, generation-level adaptive attacks, capability benchmarks, and a second model family.

AI use disclosure

Implementation and drafting were a collaboration between the author and an AI agent (Claude, Anthropic, via Claude Code), as a reference build for the Lexsi AI Safety & Alignment Hackathon; the author acknowledges this contribution. Procedures from the Scientific Agent Skills library [Kassis et al., 2026] (peer review, scientific writing, visualisation, citation checks) were used, and two independent AI reviewer agents critiqued a previous draft; the controls in §5.1 answer their main findings. All numbers are generated from result files by script.

Broader impact and ethics

The work aims to make safety repair accountable to speakers of low-resource languages. Harmful prompts come from public benchmarks; we release judgments and aggregate statistics, not new attack content. Steering artefacts can be inverted to remove refusal; we will release them only under gated access with this caveat, as we will model responses. The brittleness result is a reason to test gates against realistic inputs before deployment, not an attack recipe.

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A Pre-registration and deviations

The analysis plan (`research/00_research_plan.md`) was written before any model output and amended with dated entries as the study progressed. It is an internal, timestamped log kept by the same agent that ran the analysis, not an external registry, and its primary endpoints changed twice. The original H5 (2026-10-26) required EORS to narrow the cross-language SRR gap and to keep every language's BRR within E0 $+ \varepsilon$ at a capability cost of at most one point; the plan of 2026-10-27 18:50 made worst-language BRR against G-POOL primary, after the post hoc gate-level analyses. By the original H5, EORS (ε=.05) fails: the SRR gap widens by $+1.4$ points $[+0.0, +3.8]$, benign refusal exceeds E0 by more than five points in 2 languages (most in Swahili, $+7.5$), and capability was not measured. Deviations that affect this paper: steering strength chosen by a calibration-only rule fixed before the sweep; the end-to-end comparisons fixed before

any steering result was inspected; G-EN added as the comparator H6 names; generation moved to an A100 with same-hardware references after a hardware-agreement check; the brittleness controls and the CircuitKIT faithfulness redesign are post hoc and labelled as such. After the α=1 results, two follow-ups were pre-registered with their reading rules before generation (deviations log, 2026-10-27 23:45): the α=0.5 robustness check (Table 3) and the end-to-end greeting test (§5.4); both are scored by Qwen3Guard-4B only. After two independent AI reviews of the resulting draft, the matched-budget controls were pre-registered with research notes and a premortem (deviations log, 2026-10-28 06:40; `research/11_controls_premortem.md`): EORS-S and CPOOL-S at two budgets, two co-primary estimands with Holm correction, a gate-level validity check and a harm-versus-style check, all read by rules fixed before generation. A seed clarification was logged before any control result existed. The hardware and batching diagnostic, the correlated-error reading, the closed model's re-scoring with Qwen3Guard and the judge-gap difference-in-differences are post hoc; redrawing the calibration split and the frontier are exploratory. An independent re-review of the revised draft prompted the tier-gap reporting, routing the validity check through language identification, and the split redraws.

Table 3: **Replication at half the steering strength** (α=0.5, Tiny Aya Earth, A100, Qwen3Guard-4B). Paired changes in percentage points, 95% paired bootstrap intervals over prompt ids; the top block is against the unrepaired model (E0), the bottom block repeats the two pre-registered contrasts.

Arm	Reference	Δ mean SRR	Δ worst-language BRR	Δ mean HRR
G-POOL	vs. E0	+5.3 [+3.8, +7.0]	+21.7 [+16.0, +30.2]	+26.3 [+22.8, +29.9]
EORS ε=.05	vs. E0	+4.4 [+3.1, +6.0]	+3.8 [+0.0, +7.5]	+19.5 [+16.4, +22.8]
G-EN	vs. E0	+5.6 [+4.1, +7.4]	+28.3 [+19.8, +37.7]	+31.9 [+27.8, +36.0]
CAST-EN	vs. E0	+5.5 [+4.0, +7.2]	+22.6 [+15.1, +31.1]	+30.0 [+26.1, +34.1]
EORS ε=.05	vs. G-POOL	−0.9 [−1.3, −0.4]	−17.9 [−27.4, −12.3]	−6.8 [−8.3, −5.4]
CAST-EN	vs. G-EN	−0.1 [−0.4, +0.0]	−5.7 [−11.3, +0.0]	−1.9 [−3.0, −1.0]

B Gate-level analyses

This appendix holds the gate-level evidence behind §5.1 and §5.4: CAST against the projection gate, end to end (Figure 4) and on held-out activations (Figure 6); the pooled and per-language threshold frontier (Figure 5); gate parity and harm detection on Tiny Aya Global (Figures 7 and 8); prefix brittleness (Figure 9); and Table 1 under the second judge (Table 4).

Table 4: Table 1 under the second judge, Gemini 3.5 Flash (broad rubric); unlabelled rows dropped.

Arm	Gate / change	Δ mean SRR	Δ worst-language BRR	Δ mean HRR
G-POOL	single pooled threshold	+4.1 [+2.7, +5.6]	+30.2 [+23.6, +40.6]	+35.2 [+30.8, +39.7]
G-EN	English vector, projection gate	+4.1 [+2.8, +5.6]	+31.4 [+24.4, +42.0]	+37.1 [+32.4, +41.9]
CAST-EN	English vector, CAST gate	+3.9 [+2.6, +5.5]	+27.4 [+19.8, +35.8]	+35.2 [+30.7, +39.9]
G-CAL	per-language max-F1	+4.1 [+2.7, +5.6]	+23.8 [+17.9, +32.9]	+35.0 [+30.6, +39.5]
EORS ε=.10	per-language conformal	+4.0 [+2.6, +5.5]	+11.3 [+6.6, +17.9]	+33.0 [+28.7, +37.4]
EORS ε=.05	per-language conformal	+3.5 [+2.3, +4.9]	+7.5 [+2.8, +12.3]	+26.6 [+22.7, +30.8]
ReSta	weight arithmetic	+0.8 [−0.4, +2.0]	+17.0 [+9.4, +25.5]	+7.5 [+5.4, +9.7]
NEG	negative control	−3.2 [−4.6, −1.9]	−1.9 [−6.6, +4.7]	−2.8 [−4.8, −0.8]
B1 (no preamble)	vs. B0, Tiny Aya Global	−0.0 [−0.8, +0.7]	+3.8 [−1.9, +9.4]	+1.6 [+0.4, +2.9]

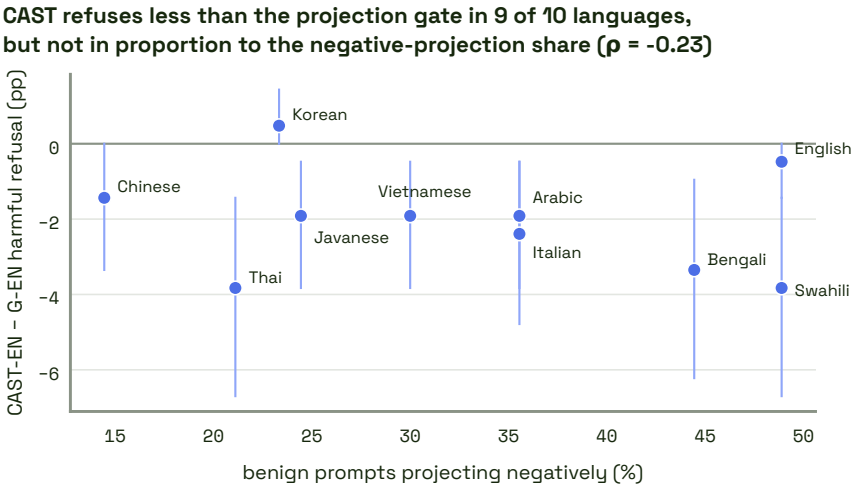


Figure 4: **CAST vs. the projection gate, end to end.** Paired per-language change in harmful-prompt refusal, CAST-EN minus G-EN (Tiny Aya Earth, Qwen3Guard-4B), against the share of benign calibration prompts projecting negatively on the condition vector (the quantity Remark 1 links to CAST's loss). Bars: 95% bootstrap over prompt ids.

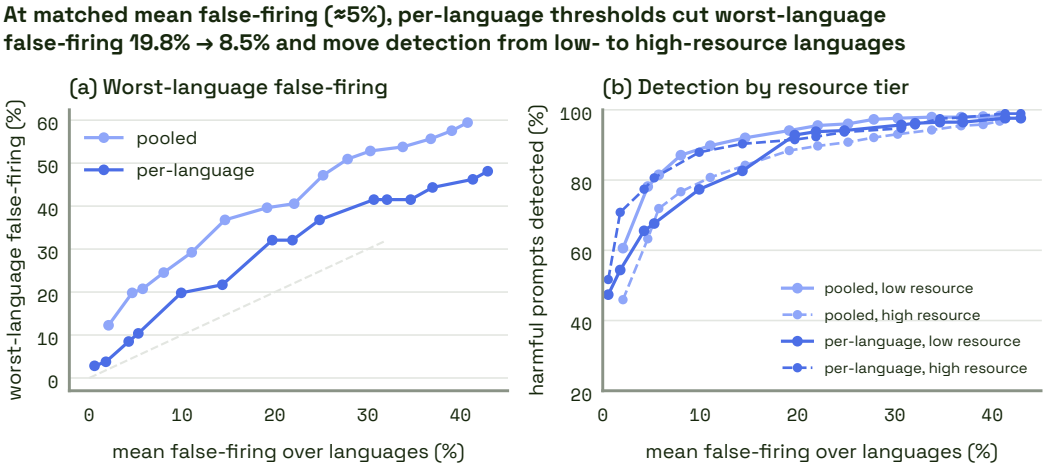


Figure 5: Gate-level frontier on Tiny Aya Earth's test activations (EORS-S direction, layer 13). For budgets $\varepsilon = .02$ to .30, thresholds from the conformal fold, pooled or per language: realised mean and worst-language false-firing on benign test prompts (a), and mean detection of harmful test prompts in the low- and high-resource tiers (b). At matched mean false-firing (4.6% pooled, 4.2% per-language), per-language thresholds cut worst-language false-firing from 19.8% to 8.5%. At similar mean false-firing (ε=.10: 9.9% vs. 11.0%), they detect 77% of low-resource harmful prompts against 90%, and 88% of high-resource ones against 81%: the redistribution of Table 2, at gate level. At larger budgets realised rates exceed ε (31% at ε=.20) because calibration and test prompts are different samples of parallel ids (§5.1). Exploratory.

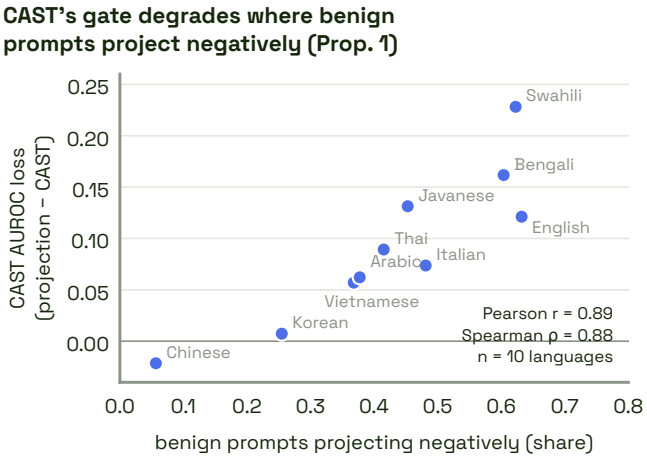


Figure 6: Gate level, Tiny Aya Global: per-language AUROC loss when gating with CAST's cosine-tanh score instead of the projection (English-fit direction, layer 14, held-out activations), against the share of benign prompts with negative projection; $n = 10$ languages, descriptive.

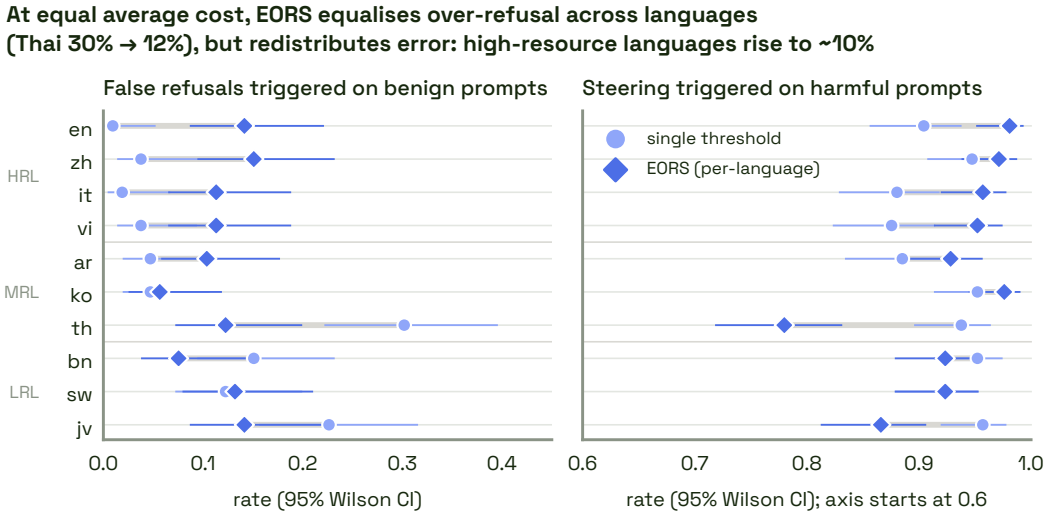


Figure 7: Gate-level parity on held-out activations (Tiny Aya Global, layer 14). At calibration-matched average cost, EORS ($\varepsilon = 0.10$) reduces the worst-language false-fire rate from 30% to 15% and raises high-resource languages to 11%–15%; post hoc. Whiskers: 95% Wilson intervals.

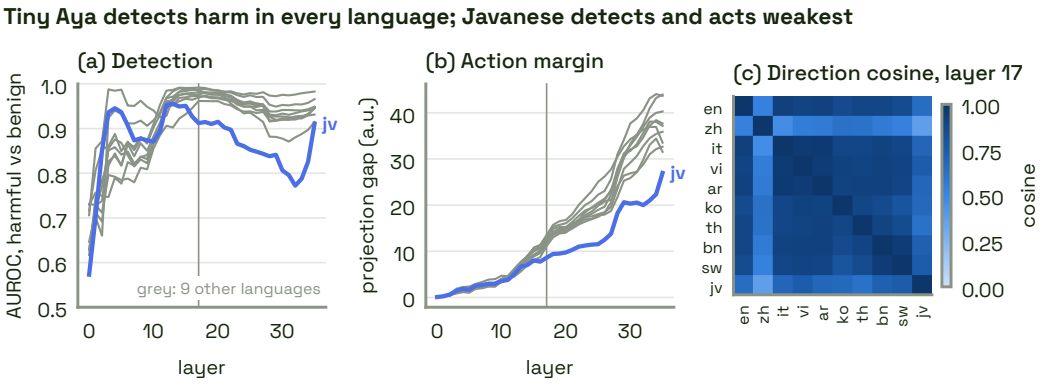


Figure 8: Tiny Aya Global, held-out prompts: (a) AUROC of the pooled harm direction by layer (best layer 17: 0.96–0.99 in nine languages, Javanese 0.91); (b) projection gap; (c) cosine between language-specific refusal directions (mean 0.79). Harm is represented in every language; the gate fires on 95% of the harmful prompts B0 answered harmfully.

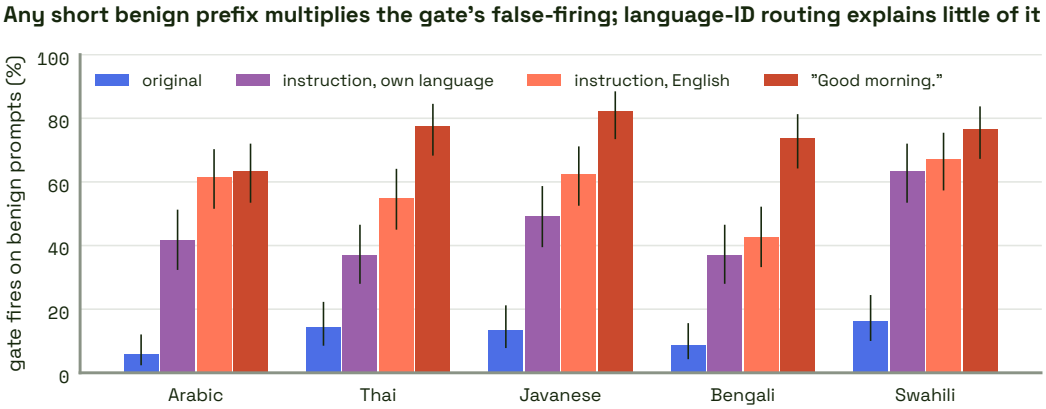


Figure 9: Gate-level false-firing of the EORS gate ($\varepsilon = 0.10$, Tiny Aya Earth) on 106 benign test prompts per language, as is or with a benign prefix; 95% Wilson intervals; constrained GlotLID picks the threshold on the modified prompt, as at inference. Post hoc.

C Baseline and natural drift

Before any repair: the baseline outcomes of Tiny Aya Global by language (Figure 10), and the natural drift from Global to its regional sibling Earth, the model every repair arm starts from (Figure 11).

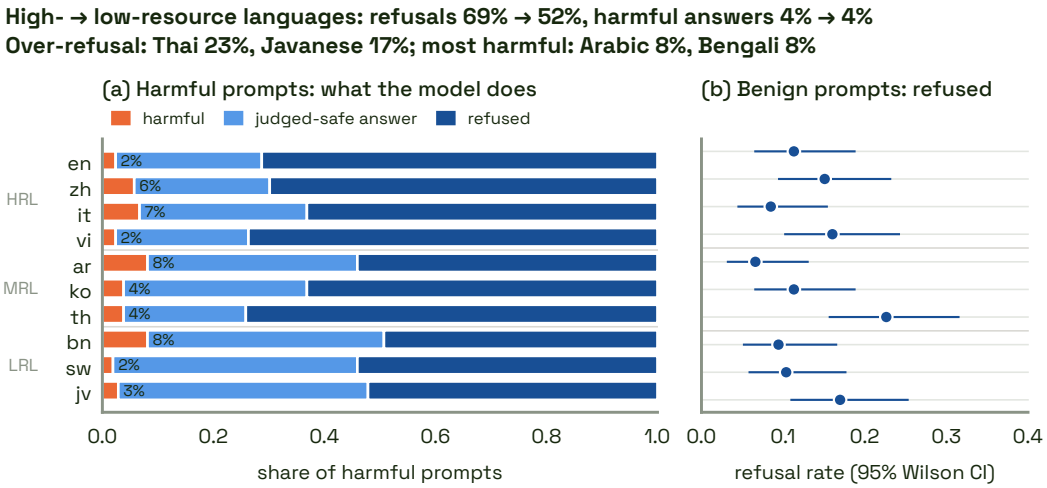


Figure 10: Baseline B0 (Tiny Aya Global): outcomes on harmful prompts and refusal of benign prompts by language, Qwen3Guard-4B; refusal falls from high- to low-resource languages (69% to 52%) while the harmful share is flat under this judge (4.3% in both tiers) and rises under Gemini (1.8% to 4.5%).

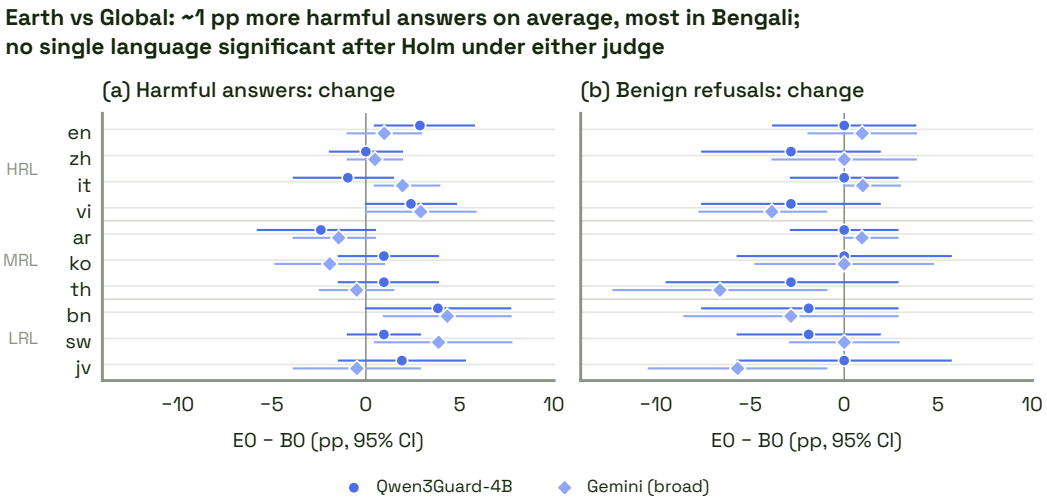


Figure 11: Natural drift: Tiny Aya Earth minus Global, paired per-language differences with 95% bootstrap intervals; circles Qwen3Guard-4B, diamonds Gemini. Exact McNemar tests, Holm-corrected; none significant.

D Circuit discovery: a negative result and tooling notes

We attempted to localise refusal with CircuitKIT's node-level EAP-IG on en, sw and bn, holding to the faithfulness standard of Hanna et al. [2024]: circuit overlap is not evidence of shared mechanism; faithfulness is. The TransformerLens port passes CircuitKIT's parity gate (maximum next-token KL 3.5×10^{-5}). Discovery needs 22.6–22.8 GiB of GPU memory at chat-template length, beyond a T4 or L4, and requires replacing TransformerLens's broadcast per-head output with an equivalent einsum. Harmful and benign prompts must be paired at equal token length, because CircuitKIT left-pads pairs and the safety preamble precedes the user turn. With length-matched pairs, held-out patching faithfulness is at most 0.06 with up to 20% of the 612 nodes, 0.41–0.47 at 40%, 0.66–0.87 at 60%, and 1.00 with all nodes (a sanity check). Our pre-fixed criterion (faithfulness ≥ 0.5 at $\leq 40\%$ of nodes) is not met, so we make no circuit claims: under this metric, refusal on these prompts is not carried by a sparse node-level circuit. SafeTune's circuit-to-layer adapter, given the top 20% of nodes, selects nearly every layer, so circuit-guided layer choice was uninformative here.

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